

# LAAS Project - J. Bahi

## **Dropout: A Simple Way to Prevent Neural Networks from Overfitting.**

*Nitish Srivastava et al., Journal of Machine Learning Research  
15 (2014) 1929-1958*

After carefully reading the research article, and without going into mathematical details, summarize it in 2-4 pages, by specifying:

- The scientific context the problem addressed.
- The related works.
- The proposed idea for solving it.
- The experimental validations of the scientific approach.
- The conclusion of the authors.
- Understanding Fundamental Concepts
  1. What is overfitting in a neural network?
  2. Why are deep neural networks particularly prone to it?
  3. What is the main idea introduced by dropout?
  4. How does dropout work in practice during training?
  5. What does it mean to “deactivate a neuron”?
  6. Which random distribution is used?
  7. What is the difference between training and test phases?
- Deep Conceptual Understanding
  1. What is co-adaptation of neurons? Why is it problematic?
  2. How does dropout prevent co-adaptation?

3. Why does dropout enforce robust feature learning?
  4. Why can dropout be seen as ensemble learning?
  5. How many subnetworks are implicitly trained?
  6. Why is this more efficient than classical bagging?
- Technical Mastery
    1. How is the dropout mask applied mathematically?
    2. Why adjust activations at test time?
    3. What is the meaning of parameter  $(p)$ ?
    4. What happens if  $(p)$  is too small? Too large?
    5. Why different values for input and hidden layers?
  - Experimental Results
    1. What gains were observed?
    2. Does dropout always improve performance?
  - Critical Thinking
    1. What mechanisms explain dropout's effectiveness?
    2. In what sense is dropout a regularization method?
    3. Can dropout be related to probabilistic methods?
    4. What are its drawbacks?
    5. When does dropout become less useful?
  - Application
    1. Where to place dropout?
    2. How to choose  $(p)$ ?
    3. When could dropout hurt performance?
    4. How adapt dropout in modern CNNs?
  - Experiments
    1. Reproduce the experimental results of Fig. 4 and Fig. 9 of the research article on the MNIST dataset using jupyter notebook.
    2. Same question with CIFAR-10 and CIFAR-100 datasets
    3. Explain your codes and discuss the results